NUMERISCHE MATHEMATIK I | BISECTION METHODS

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1. Review – Fix Point Iteration

The convergence of the Picard iteration

$$x^{k+1} = \Phi(x^k), \qquad k \in \mathbb{N}, \qquad \text{where} \qquad x^0 \in K$$

is ensured, if $\Phi : K \to K$, K is convex, and Φ is Lipschitz continues with Lipschitz constant $L_\Phi < 1$, i.e. Φ is a contraction.

Then it holds

$$\|x^{k+1} - x^k\| = \|\Phi(x^k) - \Phi(x^{k+1})\| \leq L_\Phi \|x^k - x^{k-1}\|$$

such that

$$\|x^{k+m} - x^k\| = \|\Phi(x^k) - \Phi(x^{k+1+m})\| \leq L_\Phi^m \|x^k - x^{k-1}\|, \qquad \forall m \in \mathbb{N}.$$

Hence, we have

$$\begin{aligned}
\|x^{k+p} - x^k\| &\leq \sum_{\ell=1}^{p} \|x^{k+\ell} - x^{k+\ell-1}\| \\
&\leq \sum_{\ell=1}^{p} L_\Phi^\ell \|x^k - x^{k-1}\| \\
&= L_\Phi \|x^k - x^{k-1}\| \sum_{\ell=0}^{p-1} L_\Phi^\ell \\
&\leq \frac{L_\Phi}{1 - L_\Phi} \|x^k - x^{k-1}\| \\
&\leq \frac{L_\Phi^k}{1 - L_\Phi} \|x^1 - x^0\| \to 0, k \to \infty.
\end{aligned}$$

Since the latter implies the convergence of the sequence, due to the completeness of $\mathbb{R}$ such that any Cauchy sequence converges, follows

$$\|x^* - x^k\| \leq \frac{L_\Phi^k}{1 - L_\Phi} \|x^1 - x^0\|, \quad \forall k \in \mathbb{N}$$

and ence the following a priori approximation

$$|x^* - x^k\| \leq \frac{L_\Phi^k}{1 - L_\Phi} \|x^1 - x^0\| < \varepsilon \quad \Leftrightarrow \quad k \geq \frac{\ln\left(\frac{\varepsilon(1-L_\Phi)}{\|x^1 - x^0\|}\right)}{\ln(L_\Phi)}.$$

Differently to

$$|x^* - x^k\| \leq \frac{L_\Phi}{1 - L_\Phi} \|x^k - x^{k-1}\|$$

which needs the calculation of the k-th iterate x^k.

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2. Review – Newton Iteration

As an application of the Fix Point iteration we firstly derive the following corollary

Corollary 2.1. *Let $D \subset \mathbb{R}^n$ and $\Phi : D \to \mathbb{R}^n$ a contraction on the ball $K \coloneqq \{y \in D : \|y - z\| \leq r\}$ with Lipschitz constant L_Φ with*

$$\|\Phi(z) - z\| \leq (1 - L_\Phi)r.$$

Then all three statements from Banachs Fix Point iteration are still valide.

Beweis. It suffices to show that $\Phi(K) \subset K$. It holds

$$\|\Phi(y) - z\| \leq \|\Phi(y) - \Phi(z)\| + \|\Phi(z) - z\| \leq L_\phi \underbrace{\|y - z\|}_{\leq r} + (1 - L_\Phi)r \leq r.$$

□

Let $\Phi : I \subset \mathbb{R} \to \mathbb{R}$ continuously differentiable with Fix Point $x^* \in \mathbb{R}$ and $|\Phi'(x^*)| < 1$. Then, there is for any $\varepsilon > 0$ with $|\Phi'(x^*)| + \varepsilon =: q < 1$ an $\delta > 0$ such that

$$|\Phi'(x)| \leq |\Phi(x^*)| + |\Phi(x) - \Phi(x^*)| \leq |\Phi(x^*)| + \varepsilon = q < 1$$

for any $x \in [x^* - \delta, x^* + \delta]$ and the Fix Point iteration converges.

In conclusion: There are examples, where

(1) the iteration converges (locally),
(2) diverges, or
(3) oscillates, for instance for $\Phi(x) \in \{x - x^3, x + x^3, x - x^2\}$ for the Fix Point $x^* = 0$.

For solving

$$F(x^*) = 0$$

we assume that $F : \mathbb{R} \to \mathbb{R}$ is continuously differentiable such that

$$F(x^*) = F(x) + J_F(x)(x - x^*) + o(\|x^* - x\|).$$

Then, the linear approximation is given by

$$x \mapsto F(x) + J_F(x)(x^* - x)$$

and the solution $F(x^*) = 0$ is approximated by

$$0 = F(x) + F'(x)(x^* - x) \quad \Leftrightarrow \quad x^* = x - F'(x)^{-1}F(x),$$

if $F'(x) \neq 0$ is invertible. Then the Fix Point iteration is applied to

$$\Phi_F(x) \coloneqq x - F'(x)^{-1}F(x)$$

Then the Corollary is applicable to Φ if F is twice continuously differentiable with

$$J_\Phi(x) = \frac{F(x)F''(x)}{F'(x)^2}.$$

By Taylor holds

$$\begin{aligned} |x^* - x^{k+1}| &= |x^k - \tfrac{F(x^k)}{F'(x^k)} - x^*| \\ &= \frac{|F(x^*) - F(x^k) - F'(x^k)(x^* - x^k)|}{|F'(x^k)|} \\ &= \frac{1}{2}\frac{|F''(\xi^k)|}{|F'(x^k)|}|x^k - x^*|^2 \end{aligned}$$

Furthermore, by the property if F, there is for any $\varepsilon > 0$ some $r > 0$ such that

$$|F'(x)| \geq \frac{1}{2}|F'(x^*)| \qquad \text{and} \qquad |F''(x)| \leq \varepsilon + |F''(x^*)|$$

for any $x \in \{y : |y - x^*| \leq r\}$. In summary, with the convergence of the iteration (locally for appropriate $\rho > 0$ on $K = [x^* - \rho, x^* + \rho]$) does hold

$$\begin{aligned}|x^{k+1} - x^*| &\leq \frac{|F''(x^*)| + \varepsilon}{|F'(x^*)|}|x^k - x^*|^2 \\ &= C|x^k - x^*|^2, \qquad \forall k \in \mathbb{N}.\end{aligned}$$

The convergence of the simplifies Newton iteration, i.e. the iteration on

$$\Phi(x, x^0) := x - \frac{F(x)}{F'(x^0)}$$

holds with the corollary, with the property $\Phi(x^*, x^0) = x^*$ and the continuous property around (x^*, x^*) of the derivative of the function $\Phi(\cdot, x^0) : I_\rho := [x^* - \rho, x^* + \rho] \to \mathbb{R}$, i.e.

$$\Psi(x, x^0) = 1 - \frac{F'(x)}{F'(x^0)}, \qquad \text{with} \qquad \Psi(x^*, x^*) = 0.$$

It suffices to take $\rho > 0$ small enough such that

$$\max_{x \in I_\rho} \Psi(x, x^0) < 1.$$

3. Bisection Methods

We assume for the following section

$$F : [a, b] \subset \mathbb{R} \to \mathbb{R}$$

with

$$F(a)F(b) < 0.$$

Hence, F has at least one root, i.e. there exists $x^* \in [a, b]$ such that $F(x^*) = 0$.

Generally, the bisection methods are algorithms of the following form

(1) Set $a_0 := a, b_0 := b$.
(2) For $k = 0, 1, 2, ...$ choose $t_k \in (a_k, b_k)$, calculate $f_k := f(t_k)$, and
  (a) $a_{k+1} := \begin{cases} a_k & : f(a_k)f_k < 0, \\ t_k & : \text{ else.} \end{cases}$ and
  (b) $b_{k+1} := \begin{cases} t_k & : f(a_k)f_k < 0, \\ b_k & : \text{ else.} \end{cases}$.

There are different rules for the choices of t_k.

a) Set $t_k := \frac{1}{2}(b_k - a_k)$, such that

$$|x^* - x^k| \leq \tfrac{1}{2^k}(b - a), \qquad k \in \mathbb{N}.$$

b) Set

$$t_k := a_k - f_k \frac{a_k - b_k}{f(a_k) - f(b_k)}.$$

c) Set $t_{-2} := a_0, t_{-1} := b_0$ and iteratively

$$t_k := \begin{cases} t_{k-1} - f_{k-1}\frac{t_{k-1} - t_{k-2}}{f_{k-1} - f_{k-2}} & : f_{k-1}f_{k-2} \leq 0, \\ t_{k-1} - f_{k-1}\frac{t_{k-1} - t_{k-3}}{f_{k-1} - \frac{1}{2}f_{k-3}} & : f_{k-1}f_{k-2} > 0, \quad \text{or} \quad f_{k-1}f_{k-3} \leq 0 \\ \frac{1}{2}(a_k + b_k). \end{cases}$$

For the third method (c) it holds the following: We denote the approximation for the k-th iterative t_k by

$$\varepsilon_k \coloneqq t_k - x^*,$$

where x^* is the root in an appropriate small neighborhood of the starting point in $[a, b]$. Then, the Taylor expansion yields

$$f_k = \underbrace{f^*}_{f(x^*)=0} + \sum_{\ell=1}^{\infty} c_\ell \varepsilon_k^\ell, \quad \text{where} \quad c_\ell \coloneqq \tfrac{f^{(\ell)}}{\ell!}.$$

Equivalently, we obtain for the (unmodified) update scheme

$$t_{k+1} = t_k - f_k \tfrac{t_k - t_{k-1}}{f_k - f_{k-1}} = \tfrac{t_{k-1} f_k - t_k f_{k-1}}{f_k - f_{k-1}}.$$

For the Taylor approximation, we obtain for the denominator

$$\begin{aligned} f_k - f_{k-1} &= c_1(t_k - t_{k-1}) + c_2(t_k^2 - t_{k-1}^2) + c_3(t_k^3 - t_{k-1}^3) + ... \\ &= c_1(\varepsilon_k - \varepsilon_{k-1})(1 + \mathcal{O}(\varepsilon_k + \varepsilon_{k-1})). \end{aligned}$$

For the nominator, we obtain

$$\begin{aligned} t_{k-1} f_k - t_k f_{k-1} &= (x^* + \varepsilon_{k-1})(c_1 \varepsilon_k + c_2 \varepsilon_k^2 + ...) - (x^* + \varepsilon_k)(c_1 \varepsilon_{k-1} + c_2 \varepsilon_{k-1}^2 + ...) \\ &= c_2(\varepsilon_{k-1} \varepsilon_k^2 - \varepsilon_k \varepsilon_{k-1}^2) + ... \\ &= c_2 \varepsilon_{k-1} \varepsilon_k (\varepsilon_k - \varepsilon_{k-1}) + ... \end{aligned}$$

Hence, it holds the following asymptotic behavior

$$\varepsilon_{k+1} \sim \tfrac{c_2}{c_1} \varepsilon_k \varepsilon_{k-1}$$

for a simple root. The next iteration depends on the sign of $f_{k+1} f_k$. Assuming that the unmodified iteration holds true, it holds

$$\varepsilon_{k+2} \sim \tfrac{c_2}{c_1} \varepsilon_{k+1} \varepsilon_k$$

by substituting. For the modified process, we analyse

$$\begin{aligned} 2f_k - f_{k-1} &= 2(c_1 t_k + c_2 f_k^2 + c_3 f_k^3 + ...) - (c_1 t_{k-1} + c_2 f_{k-1}^2 + c_3 f_{k-1}^3 + ...) \\ &= c_1(2t_k - t_{k-1}) + c_2(2t_k^2 - t_{k-1}) + \mathcal{O}(|t_k|^3 + |t_{k-1}|^3) \\ &= c_1(2t_k - t_{k-1})[1 + \mathcal{O}(|t_k| + |t_{k-1}|)]. \end{aligned}$$

On the other hand holds

$$\begin{aligned} 2t_{k-1} f_k - t_k f_{k-1} &= 2t_{k-1}(c_1 f_k + c_2 f_k^2 + c_3 f_k^3 + ...) - f_k(c_1 f_{k-1} + c_2 f_{k-1}^2 + c_3 f_{k-1}^3 + ...) \\ &= 2(c_1 f_{k-1} f_k + c_2 f_{k-1} f_k^2 + c_3 f_{k-1} f_k^3 + ...) - c_1 f_k f_{k-1} - c_2 f_{k-1}^2 f_k - c_3 f_{k-1}^3 f_k - ... \\ &= c_1 f_k f_{k-1} + c_2(2 f_{k-1} f_k^2 - f_{k-1}^2 f_k) + \mathcal{O}(|f_{k-1}|^3 |f_k| + |f_k|^3 |f_{k-1}|). \end{aligned}$$

Hence, it holds

$$\varepsilon_{k+2} \sim \tfrac{\varepsilon_{k+1} \varepsilon_k}{2\varepsilon_{k+1} - \varepsilon_k} \sim -\varepsilon_{k+1},$$

since $|\varepsilon_{k+1}| \gg |\varepsilon_k|$. In all different cases, we end up with two unmodified followed by a modified update scheme. Hence, it holds

$$\varepsilon_{k+2} \sim \left(\tfrac{c_2}{c_1}\right)^2 \varepsilon_k^2 \varepsilon_{k-1} \sim \left(\tfrac{c_2}{c_1}\right)^2 \varepsilon_{k-1}^3.$$

This three iterations can be performed in a single one, with error μ_k, we end up with

$$\mu_{k+1} \sim \left(\tfrac{c_2}{c_1}\right)^2 \mu_k^3.$$

Bisection-Methode